Why gdTAI V4 underperforms V2/V3 on unexposed test sets: a causal decomposition

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The observation. On the frozen common lockbox (335,479 cells; 41,931 gdT gold from BALF_BLOOD_COPD, GDT_2020AUG_woCOV cord blood, and GDTlung2023july_7p; 285,524 abT gold; 8,024 author-NK negatives; manifest SHA-256 `f997e24b…`), the development-frozen V4.3 candidate scored recall / precision / F1 = **45.6% / 96.3% / 61.9%**, versus **75.2% / 97.9% / 85.0%** for frozen V3 Round 14 on identical cells. The user has confirmed the lockbox cells were never used to *fit* V2/V3, and the project's own post-lockbox audit verified BALF was absent from V2/V3 coefficient fitting (V3's base coefficients, scaler, and intercept are numerically identical to V2). So the gap is real, not a leakage artifact. The question is what produces it.

Verdict in one paragraph. The 23-F1-point gap decomposes into four causes, in rough order of size: **(A) a threshold-transfer failure** — V4.3's operating point was estimated from early-stopped fold models and applied to a different, uncalibrated 800-tree refit whose scores saturate near 1, so a sub-percent score-scale shift on new data deletes half the positives; **(B) positive training-set narrowing** — the V4 protocol removed all sorted-γδT and silver cells from fitting, cutting positive diversity ~5×, which shows up as asymmetric Vδ1/Vδ2 recovery; **(C) feature amputation** — NK/cytotoxic context genes and the Stage-1 probability were removed to fight NK false positives, but those features carry recall signal because γδT cells *are* cytotoxic; and **(D) an asymmetric comparison frame** — V2/V3's operating points are the survivors of many adaptive rounds on related data, while V4.3 had a single honest pass. A hard residual remains after all four: even with an oracle (post-hoc, lockbox-fitted) threshold, V4.3's best achievable F1 is 0.8464 — still below V3's frozen 0.8504 — so not everything is procedural.

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1. Cause A — threshold transfer failure (dominant)

V4.3's ranking is nearly perfect; its operating point is broken:

- On BALF lockbox positives, V4.3 achieves **ROC-AUC 0.9972 / AP 0.9836** — the model *orders* γδT above αβ almost flawlessly.
- But the frozen Stage-2 threshold is **0.994098**, and the BALF positive **median score is 0.992373** — half of all true positives sit *below* the operating point. Frozen recall on BALF: 46.7%.
- The threshold was derived from grouped-OOF predictions of **early-stopped fold models**, while the deployment candidate was a **separate, uncalibrated 800-tree refit** with a different score distribution. The cutoff was estimated on one object and applied to another.
- With 800 boosting trees and no calibration, scores pile up at ≈1.0; a threshold at 0.9941 sits deep in the saturated tail, where a tiny cross-dataset score shift moves enormous cell counts across the cutoff.

This failure mode is not unique to V4.3 — it is a structural property of this modeling stack, measured independently in the gdTAI-kimi experiments: BALF positive medians of 0.963–0.969 against transferred thresholds of 0.95–0.99, and inner-calibrated FPR targets missing by up to 19× when transferred toward GSE144469 (`gdtai_kimi_report.md` §8.2). XGBoost/HistGBM scores without deployment-matched calibration plus tail-quantile thresholds will keep producing exactly this failure.

Fixable? Yes, procedurally: derive OOF predictions from full-length fold refits of the *same* architecture as the deployment model, calibrate on those OOF scores (Platt/isotonic), and never transfer a threshold across model objects. This alone recovers most of the recall gap (V4.3's own threshold sweep reaches recall ≈ V3's at FPR 0.537% — too dirty to deploy, but it proves the scores carry the information).

2. Cause B — positive training-set narrowing (real model-content gap)

- The V4 protocol (v1.2) removed **all sorted-γδT cells and all silver cells** from both fitting stages. Sorted cohorts supplied 48,879 of V3's 57,776 gold positives (85%). V4.3's positive class was therefore much smaller and much less diverse in V-gene usage, tissue, and chemistry.
- The measurable signature: V4.3 recovers only **43.2% of productive-TRDV2 BALF cells**, versus 92.7% for V2 high-F1 — it learned Vδ1-biased receptor rules from a narrowed positive set.
- 389 of V4.3's 454 BALF false negatives are V2 true positives with strong TCR-UMI support (median 136 UMIs vs 199.5 for V4.3 TPs) — these are real, high-quality γδT cells the narrowed model never learned, not dropout junk.
- Notably, the lockbox's two sorted cohorts (cord blood, GDTlung) are *novel biology* to V4.3 (it never trained on sorted-style cells at all), while V3's operating point was explicitly validated for cord-blood recovery during the Round 12-vs-14 decision. Source recall deficits: BALF −45%, GDTlung −37%, cord blood −28%.

Fixable? Partially, by protocol amendment: sorted cells can return to fitting at low weight (kimi round 1 used weight-0.25 sorted positives without NK damage — it was TRD-only *silver* positives that imported NK contamination, and v1.2 was right to remove those). Whether this closes the Vδ2 gap can only be tested in a new precommitted run, not on the consumed lockbox.

3. Cause C — feature amputation

V4.3's Stage 2 uses 153 TCR genes + CD3D/E/G + CD4 + FOXP3. To fight NK false positives, the NK/cytotoxic context genes (KLRD1, NKG7, GNLY, FCER1G, TYROBP) were excluded from both stages, and the Stage-1 probability is not passed to Stage 2. But γδT cells are cytotoxic cells — these features carry *positive-class* signal, not just NK signal. The kimi rounds used the 197-gene space (including these genes) plus the Stage-1 OOF probability, and its HistGBM hit held-out recall 0.83–0.90 on HRA005041/GSE144469 — with the NK problem controlled by *labels* (assayed-sublibrary NK negatives), not by feature removal. Removing features to fix a label problem threw away recall signal.

4. Cause D — an asymmetric comparison frame (even with unexposed cells)

“Not used in fitting” is necessary but not sufficient for a level comparison:

- V3’s *promotion* was decided partly on cord-blood and BALF behavior (Round 12 vs 14 was selected on mean F1 across frames including these cohorts); V2/V3’s thresholds are the survivors of ~38 candidate rounds plus six candidate lanes of adaptive iteration on related data. V4.3 had exactly one honest pass.
- The lockbox’s positive sources are sorted/simulated-positivity cohorts on which V3’s threshold was, in effect, historically validated. V4.3’s threshold never saw them.
- The converse is also documented: V4.3’s apparent purity edge is illusory — it has *higher* paired-αβ FPR than V3 in 7 of 8 extension datasets (all seven McNemar-significant after BH correction). Nothing about the frame was rigged in V4.3’s favor either.

5. The residual that is real

Even granting V4.3 an oracle threshold fitted on the lockbox itself (diagnostic only, not deployable), its maximum F1 is **0.8464 < 0.8504** (V3 frozen). So after removing all procedural causes, the V4.3 model *content* is still marginally worse than V3 on this lockbox — consistent with causes B and C being genuine capability losses, not just calibration artifacts.

6. What was excluded as a cause (project’s own audits, verified here)

- Stage-1 gate: passes 98.2–99.5% of locked positives — not the bottleneck.
- CD4/Treg exclusions: ≤1.41% effect.
- Feature-matrix integrity: 59,924,265 values bit-exact against the frozen training cache.
- FN cell quality: FNs have strong TCR-UMI support (median 136).
- NK label quality: repaired in the V4.2 line (latent NK reference; 469 confident NK, strict receptor/adaptor expression rule) — and, independently, kimi round 3 showed the historical NK-FPR inflation was mostly censoring-contaminated NK labels, fixed by assayed-sublibrary restriction.

7. Implications

1. **V3 Round 14 remains the correct default.** The lockbox is consumed; no further model selection on these cohorts is legitimate.
2. **If a V4.5 is ever precommitted**, the three cheap fixes with expected large effect:
    1. thresholds from OOF predictions of full-length fold refits + calibrator fit on those OOF scores, applied to the identically-trained deployment object; (b) restore sorted-cell positives at low weight (keep silver out); (c) restore cytotoxic context features and control NK through labels (assayed-sublibrary NK negatives, latent NK reference) plus an abstain zone — not feature removal.

3. **Interpretation guardrail for the program:** the honest single-pass pipeline “costs” ~5–10 F1 points relative to the adaptively-developed incumbent. That delta is the measured price of V2/V3’s hidden adaptive iteration and leakage history — it should be the expected baseline gap when judging any future honestly-developed model, not read as V4 being uniquely broken.

Appendix. Key artifacts relied on

- `Integrated_dataset/logs/gdT_prediction/gdtai_v4_3_common_lockbox/evaluation_summary.json` (lockbox metrics, promotion decision)
- `Integrated_dataset/logs/gdT_prediction/gdtai_v4_3_recall_failure/summary.json` (AUC/AP, positive median vs threshold, TRDV2 asymmetry, FN UMI support, V2≡V3 base equality, extension FPR McNemar results)
- `Integrated_dataset/logs/gdT_prediction/gdtai_v4_3_final_evaluation/gdtai_v4_final_conclusion.md` (post-hoc frontier 0.8464 < 0.8504)
- `Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_nk_definition_repair/summary.json`
- `Integrated_dataset/logs/gdT_prediction/gdtai_kimi/gdtai_kimi_report.md` (rounds 1–3: threshold-transfer replication, sorted/silver ablation, NK label repair)